

Cicatricial Eyelid Closure Following Severe Eyelid Blistering in Stevens–Johnson Syndrome: A Case Report

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ABSTRACT

Stevens–Johnson Syndrome (SJS) is a rare but severe immune-mediated mucocutaneous reaction, most commonly triggered by medications. Ocular involvement occurs in a significant proportion of cases and may progress to sight-threatening cicatricial complications. We report a case of a 7-year-old female who developed extensive systemic manifestations of SJS following exposure to a sulphonamide-containing drug. The patient presented with persistent fever, mucosal bleeding, generalized erythematous rashes, and blistering involving the eyes, oral cavity, genitalia, and skin. Ophthalmic examination revealed severe eyelid blistering with necrosis, haemorrhagic crusting, cicatricial eyelid closure, conjunctival epithelial sloughing, and early symblepharon formation. Urgent ophthalmologic intervention included frequent ocular lubrication, broad-spectrum topical antibiotics, cycloplegics, and daily mechanical separation of eyelid adhesions with a glass rod lubricated with steroid/antibiotic ointment. These measures prevented permanent cicatricial deformities and preserved cornea integrity. Laboratory findings revealed moderate anaemia, electrolyte imbalance, and hypoalbuminaemia. A multidisciplinary approach involving Paediatrics, Otorhinolaryngology, Dermatology, Ophthalmology, Dietetics, and Plastic Surgery was instituted. Supportive care, systemic corticosteroids, and cyclosporine therapy resulted in progressive clinical improvement within one week, with restoration of visual acuity to 6/6 bilaterally. This case highlights the importance of early recognition, prompt withdrawal of the offending agent, and urgent ophthalmologic intervention to prevent permanent ocular sequelae in paediatric Stevens–Johnson Syndrome.

Keywords: *Stevens–Johnson Syndrome, cicatricial eyelid closure, sulphonamide, ocular complications, paediatric dermatology, case report*

INTRODUCTION

Stevens–Johnson Syndrome (SJS) is a severe immune-mediated hypersensitivity reaction characterized by widespread epidermal necrosis and mucosal involvement.¹ It is most commonly drug-induced, with sulphonamides, anticonvulsants, and non-steroidal anti-inflammatory drugs among the leading triggers.²

Although rare, SJS carries significant morbidity and mortality, particularly in children.

Ocular involvement is reported in up to 50–80 percent of cases and may range from mild conjunctivitis to severe keratitis, symblepharon formation, and cicatricial eyelid deformities.^{3,4} Intense inflammation and blistering of the eyelids can lead to fibrosis, ankyloblepharon, and functional eyelid closure if not promptly managed.^{4,5} These complications may result in exposure keratopathy, cornea scarring, and permanent visual impairment.

Early diagnosis, immediate withdrawal of the offending drug, and multidisciplinary management are critical to improving outcomes and preventing long-term disability. This case underscores the

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severity of ocular involvement in paediatric Stevens-Johnson syndrome and highlights the critical role of prompt ophthalmic care in mitigating irreversible visual impairment from intense inflammatory sequelae. Furthermore, it reinforces the necessity of a comprehensive, multidisciplinary approach to ensure holistic management and improved long term outcome.

CASE PRESENTATION

A 7-year-old female was brought to the Children's Emergency Unit with a one-week history of persistent high-grade fever associated with bleeding from the mouth, eyes, and genitalia. The illness began shortly after administration of a sulphonamide-containing medication (*Septin-sulfamethoxazole/trimethoprim*) at a peripheral hospital.

Within days, she developed generalized painful erythematous rashes that progressed to blistering across multiple mucocutaneous sites. She experienced painful micturition, odynophagia, and progressive inability to open both eyes due to marked eyelid swelling and blister formation. Ocular symptoms were severe, with crusting, necrosis, and fusion of the eyelid margins. Additional symptoms included productive cough with yellowish sputum, intermittent non-bilious vomiting, reduced urine output, constipation, and painful genital and buttock blisters.

General examination revealed an acutely ill looking and dehydrated child. She was febrile, and in significant painful distress, acyanosed and anicteric.

A generalized skin and facial hyperpigmented and hypopigmented macules and papules. The targetoid lesions typically depicts Steven-Johnson syndrome lesion. Mucosal involvement include; severe

haemorrhagic crusting of the Lips, with sloughing.

Erosion of the Oral mucosa. Oral examination showed mucosal ulceration and bleeding.

The pulse rate and respiratory rate were normal. Bilateral rhonchi and Lung basal crepitations were noted. The abdomen was full, moves with respiration, no palpable mass or organomegaly.

Ocular Examination

At presentation the visual acuity was counting fingers in both eyes taking by the bedsides with difficulty in parting the lids. Both eyelids were markedly swollen, with extensive blistering, necrotic tissue at the lid margins, and inflammatory crusting. The eyelids were effectively sealed shut due to exudates and evolving cicatricial adhesions, preventing spontaneous eye opening. Ulcerations and necrosis of eyelid skin and haemorrhagic crusting along the lid margins was noted.

Marked conjunctival injection with marked epithelial sloughing and symblepharon formation was noted in both eyes. The cornea was clear, anterior chamber depth normal. Funduscopy was deferred at first review but normal fundi were noted at subsequent reviews.

Laboratory findings demonstrated moderate anaemia, hypoalbuminaemia, hypokalaemia, and mild uraemia. Liver function tests were within normal limits.

Diagnosis

A diagnosis of Stevens-Johnson Syndrome was made based on characteristic mucocutaneous findings and the temporal association with sulphonamide exposure.^{1,2} v Severe eyelid necrosis and inflammatory adhesion raised concern for cicatricial eyelid closure and ankyloblepharon formation.^{4,5}



Figure 1: At presentation showing severe eyelid blistering, necrotic margins, and eyelid closure secondary to mucocutaneous involvement in Stevens–Johnson Syndrome.

Written informed consent for publication of this case and accompanying image was obtained from both parents of the child.

Management

The offending drug was immediately discontinued. A multidisciplinary team comprising of Paediatrics, Dermatology, Ophthalmology, Otorhinolaryngology, Dietetics, and plastic surgery specialists was involved.

Supportive management included barrier nursing, intravenous fluid resuscitation with normal saline and dextrose saline, correction of electrolyte imbalance, and daily wound care using povidone iodine

and saline dressings with silver sulfadiazine. Bilateral aural toileting was done and Otomed ear drops for bilateral Otitis Externa. Systemic therapy included intravenous methylprednisolone followed by hydrocortisone. Oral cyclosporine was administered as adjunct immunomodulatory therapy. Evidence suggests that early cyclosporine may reduce disease progression by inhibiting T-cell mediated epidermal apoptosis.⁶

Additional supportive treatments included analgesia, antiseptic mouthwash, zinc supplementation, multivitamins, and nutritional rehabilitation. Intravenous antibiotics were administered to take care of

infections. Wound debridement was done under Local anaesthesia. And child was transfused a pint of fresh whole blood to correct the anaemia and continuous loss from the wounds.

Ophthalmologic management included frequent ocular lubrication with artificial tears hourly and ointment antibiotics at night. Broad spectrum topical antibiotics(moxifloxacin) instillation qds to

prevent infection and daily sweeping of the fornices with glass rod lubricated with steroid/ antibiotic ointment under local anaesthesia to prevent eyelid adhesion and symblepharon formation, consistent with recommended early ocular interventions.^{3,4} Daily instillation of cycloplegics (Atropine) to reduce pain and prevent synechiae. Weekly fluorescein staining of the Cornea.



Figure 2: Second day after wound debridement and glass rodding with steroid and antibiotic ointment under local anesthesia to prevent cicatricial adhesions.

Outcome

Within one week of initiating treatment, the patient's visual acuity was 6/6 in both eyes. Slit lamp examination revealed that the cornea in both eyes did not stain with fluorescein dye, indicating intact epithelial integrity. Extraocular motility was initially restricted to about 50% in all directions of gaze in both eyes during the first three days of treatment, but subsequently improved to full motility in all directions of gaze by the end of the first week. These findings confirmed

significant recovery of ocular function following early ophthalmologic and multi-disciplinary intervention.

The patient showed progressive clinical improvement. Cutaneous lesions began to re-epithelialize, mucosal bleeding subsided, and systemic symptoms resolved. Ocular inflammation reduced significantly following ophthalmologic intervention, although close follow-up was maintained to monitor for late cicatricial sequelae.



Figure 3: One week following multidisciplinary treatment showing marked reduction in inflammation and improvement in eyelid mobility.

DISCUSSION

Stevens–Johnson Syndrome is a life-threatening immune-mediated reaction characterised by widespread keratinocyte apoptosis triggered by drug hypersensitivity.¹ Sulphonamides remain among the most well-established causative agents.²

Ocular involvement is common and may be severe. Acute eyelid blistering and conjunctival inflammation can progress to fibrosis and cicatricial deformities if inadequately treated.^{3,4} Ankyloblepharon and cicatricial eyelid closure occur as a result of inflammatory adhesion between opposing epithelial surfaces during healing.^{4,5} Early ophthalmologic intervention, including lubrication, topical antibiotics, corticosteroids, and mechanical separation of lid adhesions, is essential to prevent permanent scarring.³

Management principles of SJS mirror burn care protocols, emphasizing fluid resuscitation, infection prevention, and nutritional support.¹ The use of systemic corticosteroids and cyclosporine remains debated; however, some studies suggest that cyclosporine may reduce mortality and halt

disease progression by suppressing immune-mediated keratinocyte apoptosis.⁶

CONCLUSION

This case illustrates the severe systemic and ocular manifestations of Stevens–Johnson Syndrome in a child following sulphonamide exposure. The development of severe eyelid blistering with cicatricial closure highlights the risk of sight-threatening complications. Prompt recognition, withdrawal of the offending agent, intensive supportive care, and early ophthalmologic intervention are essential to improve outcomes and minimize long-term morbidity. This case underscores the critical importance of early multidisciplinary care and aggressive ophthalmologic management in preventing long-term visual impairment in paediatric SJS.

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